Project Phase 7: Optimizing with Serverless - Smart & Efficient

Due Date: End of Week 10 **Points:** 100

Scenario

The Riverside Cafe owners now need a daily sales report emailed to them. The team's first attempt, running the report on the main web servers, slowed down the website. Their second idea, running a dedicated server just for the report, is too expensive for a task that only runs for a few minutes a day. They need a better way.

Your task is to design and implement a modern, cost-effective, serverless architecture to solve this reporting problem.

Project Goals

1. **Decouple the Reporting Task.** Isolate the report generation logic from the primary web application to ensure customer performance is never compromised.
2. Use a Function-as-a-Service (FaaS) offering to run the report-generation code on-demand, eliminating the need for a dedicated server.
3. **Automate with Event-Driven Triggers.** Implement a scheduler that automatically triggers the reporting process at a set time each day.
4. **Deliver the Report.** Use a notification service to email the final report to stakeholders.

Technical Requirements

- The compute function must run without a provisioned server (e.g., using AWS Lambda).
- The function must be placed within your VPC to securely access the production RDS database.
- The function must be triggered automatically on a daily schedule.
- You will be provided with the Python code for the function (`report-generator.zip`).

Submission Instructions

You will submit a single PDF document to Moodle. This document must contain all parts of the deliverable outlined below.

Deliverable Breakdown

- **Problem Analysis & Solution Design:** Explain the two main drawbacks of the initial ideas (running on existing web servers vs. a new dedicated server). Explain why a serverless FaaS approach is the ideal solution for this problem, focusing on **performance isolation** and **cost-effectiveness**.

- **The Serverless Architecture:** Provide a detailed diagram of your serverless workflow. It must show the scheduled trigger, the Lambda function, its integration into your VPC, the RDS Database, the Notification Service (e.g., SNS), and the final email delivery.
- **VPC Integration & Security:** Explain why the Lambda function needs to be associated with your VPC. Detail the security group configuration you created to allow the Lambda function to securely connect to the RDS database. Provide a screenshot of your Lambda function's final configuration (runtime, handler, VPC settings, etc.).
- **Automation and Verification:** What service did you use to schedule the daily execution? Provide the "cron expression" you used and explain what it means. Describe how you tested the complete workflow. Provide a screenshot of the final sales report email you received after a successful test run.

Grading Rubric

Criteria	Description	Points
Analysis & Solution	The analysis of the problem is accurate, and the justification for a serverless solution is strong and well-articulated.	25
Architecture Diagram	The diagram is clear, accurate, and includes all required components of the serverless workflow.	20
VPC & Security	The need for VPC integration is understood. The security group configuration is correct and well-explained.	25
Automation & Verification	The scheduling mechanism is correctly configured and explained. Screenshot evidence proves the entire workflow functions correctly.	30
Total		**100**